

Module 02 — Project Setup, Repo Structure & Execution Sequence

Tier: ● Beginner · Duration: 60 min · Prerequisites: Module 01

Change from original plan: The execution sequence (parse → compile → execute → report) was previously a standalone 60-minute module. It's absorbed here as a 15-minute block at the end of setup, where it belongs conceptually. Running `dbt run` is impossible to understand without knowing what it actually does.

Agenda

Time	Duration	Topic	Learning Goal	Mode	Participant Activity	Materials	Trainer Notes	Checkpoint
00:00	10 min	Recap Module 01	Confirm mental model before introducing new content	Q&A	Answer from memory, no notes	—	Ask all 4 prep questions cold. Stop and revisit any wrong answer before continuing.	All 4 correct
00:10	15 min	<code>profiles.yml</code> anatomy	Understand how dbt connects to Snowflake	Present + live file	Read along in their own clone	<code>profiles.yml</code> from repo	Walk: <code>target</code> , <code>outputs</code> , <code>account</code> , <code>role</code> , <code>warehouse</code> , <code>schema</code> . Explain dev vs. prod target split. Key point: dev schema = <code>TESTING.dev_{name}</code> — never writes to Silver or Gold.	"What schema does your dev target write to?"
00:25	15 min	<code>dbt_project.yml</code> deep-dive	Know what the project config controls	Present + live file	Annotate own copy	<code>dbt_project.yml</code>	Cover: <code>name</code> , <code>model-paths</code> , <code>+materialized</code> per layer, <code>+tags</code> , <code>+database</code> , <code>+schema</code> . Show how Silver inherits <code>materialized: table</code> from config and how <code>+database</code> enforces layer separation.	"Where is the default materialization for Gold models set?"
00:40	10 min	CLI command reference	Know the core commands and when to use each	Present	Write down command table	This doc	Don't demo all commands — just walk the table. You'll use them hands-on in Module 06.	"What command runs models AND tests together?"

Time	Duration	Topic	Learning Goal	Mode	Participant Activity	Materials	Trainer Notes	Checkpoint
00:50	10 min	The execution sequence	Understand what happens when you run dbt run	Present	Annotate diagram	Whiteboard	Five phases: Parse → Resolve → Compile → Execute → Report. Which phase produces which error type. One diagram, no more.	"At which phase does a Jinja syntax error appear?"
01:00	—	Session close + prep questions	—	—	—	—	No exercise this session — hands-on begins in Module 03 when you write your first Jinja.	—

Content

Part A — profiles.yml : How dbt Connects to Snowflake

profiles.yml is NOT stored in the dbt project repo. It lives in ~/.dbt/profiles.yml on your machine. It contains credentials — never commit it to git.

```
analytics:
  target: dev          # which output to use by default
  outputs:
    dev:
      type: snowflake
      account: abc123.eu-west-1
      user: jane@company.com
      authenticator: externalbrowser # SSO login
      role: analytics_service_role
      warehouse: COMPUTE_WH_DEV
      database: SILVER_DEV
      schema: TESTING__dev_jane      # your personal dev schema
      threads: 4

  prod:
    type: snowflake
    account: abc123.eu-west-1
    user: dbt_prod@company.com
    authenticator: snowflake
    role: analytics_service_role
    warehouse: COMPUTE_WH_PROD
    database: SILVER
    schema: PUBLIC                  # prod target; your orchestrator uses this
    threads: 8
```

Critical rule: Your dev target schema is TESTING.dev_{yourname} . You can break anything there. You can't write to SILVER or GOLD from your laptop — that's the orchestrator's job.

Test your connection:

```
dbt debug  # checks profiles.yml, project config, Snowflake connectivity
```

If this passes, you're ready to run models.

Part B — dbt_project.yml : The Project Config

```
name: analytics
version: "1.0.0"
profile: analytics      # must match the key in profiles.yml

model-paths: ["models"]
test-paths: ["tests"]
macro-paths: ["macros"]
seed-paths: ["seeds"]

models:
  analytics:
    staging:
      +materialized: view  # all staging models default to view
      +tags: ["staging"]

    silver:
      +materialized: table  # Silver defaults to table
      +tags: ["silver"]
      +persist_docs:
        relation: true
        columns: true      # writes dbt descriptions to Snowflake column comments

    gold:
      +materialized: table
      +tags: ["gold"]
      +persist_docs:
        relation: true
        columns: true
```

The `+` prefix means the config applies to all models in that folder and subfolders. Individual models can override it with a `{{ config() }}` block at the top of the SQL file.

Why `models: analytics:` — the project namespace

The `analytics:` key under `models:` is the project name namespace. It must match `name: analytics` at the top of the file. Its job is to scope your config to *your* models only — not to models from installed packages.

If you install a package like `dbt_utils`, it brings its own models into the project. Without the namespace, any config you write under `models:` would apply to package models too. With it, package models stay isolated under their own namespace:

```
models:
  analytics:      # your models – configs apply here only
```

```
silver:
  +materialized: table

dbt_utils:      # an installed package – separate namespace
+materialized: view
```

Can you skip it? Technically yes — dbt still works. But if you ever install a package, your configs could bleed into its models unexpectedly. Always include it.

Key decisions encoded in this file:

- Staging is always a view — never a table
- Silver and Gold persist docs to Snowflake so Power BI and Cortex AI can read column descriptions
- Tags enable selective runs: `dbt build --select tag:silver`

+database and +schema — routing layers to the right Snowflake location

+materialized controls *how* dbt builds a model. +database and +schema control *where* it lands. Without them, every model writes to whichever database and schema `profiles.yml` specifies — the same location for every layer. With them, you can route Silver and Gold to separate Snowflake databases directly from the project config:

```
models:
  analytics:
    silver:
      +materialized: table
      +database: SILVER    # Silver models always write to this database
      +schema: PUBLIC

    gold:
      +materialized: table
      +database: GOLD      # Gold models write to a separate database
      +schema: PUBLIC
```

The layer separation visible in Snowflake (`SILVER` vs `GOLD` databases) is enforced here — not by convention, not by the profile, but by this config.

The dev/prod problem: Hardcoding `GOLD` writes to the production `GOLD` database even on a dev run. That breaks environment isolation. The fix is a `generate_database_name` macro that switches the value based on `target.name` . That's covered in the Intermediate tier. For now: understand what `+database` does, and know that hardcoding without the macro is dangerous in practice.

Part C — Core CLI Commands

Command	What it does	When to use
<code>dbt debug</code>	Validates connection and config	First thing after cloning the repo
<code>dbt compile</code>	Renders Jinja → raw SQL without running	Inspecting what SQL dbt will execute
<code>dbt run</code>	Executes models (no tests)	Never in CI; use <code>dbt build</code>
<code>dbt test</code>	Runs tests only	Debugging a specific failing test
<code>dbt build</code>	Runs models + tests together	Always in CI; preferred locally too

Command	What it does	When to use
<code>dbt docs generate</code>	Builds documentation site	Before <code>dbt docs serve</code>
<code>dbt docs serve</code>	Serves docs locally at localhost:8080	Browsing the DAG and model docs
<code>dbt ls</code>	Lists all models matching a selector	Checking what a selector targets
<code>dbt run --select dim_patient+</code>	Runs <code>dim_patient</code> and all downstream models	Targeted rebuild after a change

The most important distinction: `dbt run` does NOT run tests. Use `dbt build`.

Part D — The Execution Sequence

When you run `dbt run` or `dbt build`, five phases execute in order:

1. PARSE

Read all .sql and .yaml files, validate Jinja syntax
↳ Fails here: Jinja syntax errors, missing macro definitions

2. RESOLVE

Build the DAG – resolve all {{ ref() }} and {{ source() }} calls
↳ Fails here: circular references, missing models

3. COMPILE

Render Jinja → raw SQL, write to target/compiled/
↳ Fails here: undefined variables, bad config blocks

4. EXECUTE

Send compiled SQL to Snowflake
↳ Fails here: SQL errors, permission errors, data type mismatches

5. REPORT

Log results, write artifacts (manifest.json, run_results.json)
↳ Always runs; even failed runs produce a report

flowchart LR

P["1. PARSE
Read .sql + .yaml
Validate Jinja"]

R["2. RESOLVE
Build DAG
Resolve ref() source()"]

C["3. COMPILE
Jinja to SQL
Write target/compiled/"]

E["4. EXECUTE
Send SQL
to Snowflake"]

X["5. REPORT
Log results
Write artifacts"]

P --> R --> C --> E --> X

EP["Jinja syntax errors
missing macros"]

ER["circular refs
missing models"]

EC["undefined vars
bad config"]

EE["SQL errors
type mismatches"]

EX["always runs"]

P -. -> EP

R -. -> ER

C -. -> EC

E -. -> EE

X -. -> EX

```

classDef phase fill:#f8fafc,stroke:#64748b,color:#1e293b
classDef fail fill:#fef2f2,stroke:#fca5a5,color:#991b1b
classDef pass fill:#f0fdf4,stroke:#86efac,color:#166534
class P,R,C,E,X phase
class EP,ER,EC,EE fail
class EX pass

```

Why this matters: When you get an error, the phase tells you where to look.

- **Compilation Error** → your Jinja is wrong → check the template, not the SQL
- **Database Error** → Snowflake rejected the SQL → check `target/compiled/` to see the actual SQL that ran
- **Dependency Error** → a `ref()` points to a model that doesn't exist → check spelling and file path

The compiled SQL lives here:

```

target/
  compiled/
    analytics/
      models/
        silver/
          dim_patient.sql ← this is what dbt actually sent to Snowflake

```

Always inspect `target/compiled/` when debugging a failing model.

Reference Material

- [dbt profiles.yml docs](#)
- [dbt project.yml reference](#)
- [dbt CLI commands](#)

Other project files you'll encounter in the repo

These don't need deep coverage now, but you'll see them in the project:

- **packages.yml** — declares external dbt packages (e.g. `dbt_utils`). Run `dbt deps` to install them into `dbt_packages/`. We use `dbt_utils` for surrogate key generation.
- **seeds/** — CSV files for small, static lookup tables that don't exist in a source system (e.g. a list of excluded test accounts, or a country-code mapping). Run `dbt seed` to load them. Don't use seeds for data that changes frequently.
- **analyses/** — SQL files that use `ref()` and `source()` for lineage tracking, but are never materialised. Useful for audit queries during a migration (e.g. "does our new Silver model match the old stored procedure?") without polluting the production model set.

Prep Questions for Module 03

1. Where does `profiles.yml` live — inside the repo or outside it? Why?
2. What does `dbt build` do that `dbt run` does not?
3. At which execution phase does a Jinja syntax error appear?

4. Where can you find the compiled SQL that dbt sent to Snowflake?